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RESEARCH-ARTICLE

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The Design and Application of Multimodal Immersive Games for C-STEAM Programs

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Abstract

Most of the current C-STEAM programs rely on traditional lecture methods for situational awareness, which can convey knowledge but has limitations in stimulating students' interest and promoting active exploration and interaction. Experiential learning can effectively promote learners' situational awareness by simulating real task situations, enabling them to understand and apply knowledge more deeply. However, current experiential educational games are still characterized by a lack of interactivity and fun, as well as a lack of cultural immersion. In this study, we designed and developed a multimodal immersive game for C-STEAM courses and applied it to the C-STEAM program. By analyzing questionnaire scales and interviewing students, it was confirmed that the game could promote students' perception of cultural situations, deepen their understanding of cultural connotation knowledge, and stimulate their interest in subsequent cultural learning.

CCS Concepts

• **Software and its engineering** → Software creation and management; Designing software; Software design engineering.

Keywords

C-STEAM, experiential learning, multimodal immersive games

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1 Introduction

With the intensification of economic globalization and international competition, STEM education has been vigorously developed in various countries due to its potential value in cultivating students' innovation ability. In the past decade, China has made some progress in exploring the practical path of STEM education in response to the needs of the state and society, but there are some problems, such as focusing on science but lacking the integration of humanities [8]. To explore the localization path of STEAM education, Zhan et al. proposed cross-disciplinary cultural heritage education (C-STEAM) [20]. C-STEAM, which highly integrates cross-disciplinary knowledge in the context of traditional culture, improves students' humanistic literacy and cultural identity and cultivates students' scientific spirit. Studies have been conducted to offer a series of C-STEAM featured programs in conjunction with traditional Chinese culture, such as an intelligent wooden arch bridge program based on mortise and tenon culture [17], a cultural sandbox program incorporating landmarks in Guangzhou [4], a dragon boat program related to the culture of the Dragon Boat Festival [19], and so on.

Contextual cognition theory suggests that all mental activities of an individual are dependent on the environment and that cognitive activity involves two interacting, intrinsically dialectical processes, i.e., exogenous processing (i.e., the process of adaptation), which is dependent on the environmental stimuli, and intrinsic processing (i.e., the process of interpretation and construction), which is dependent on the person's pre-existing subjectivity [16]. Due to the rich situational attributes in the process of cultural inheritance, C-STEAM education should also be a series of thinking and practicing activities rich in traditional cultural contexts. Through the creation of real situations or simulated situations that are as close as possible to real-world patterns, it stimulates students' embodied cognition of culture, guides them to have multi-sensory experiences, building up a sense of cultural immersion and penetrating them from the outside to the inside, ultimately facilitating their understanding of cultural connotations [9].

Experiential learning is a style of learning that emphasizes the acquisition of knowledge, skills, and attitudes by learners through hands-on activities and experiences in real-world situations. By simulating real-life task situations and providing multi-sensory interactive feedback [14], this type of learning effectively promotes situational awareness and enables learners to understand and apply knowledge in a deeper way. Therefore, multimodal interaction technology can play an important role in experiential learning by integrating visual, auditory, tactile, and other multi-sensory information to provide learners with richer and more realistic learning contexts [13]. As an innovative way of experiential learning, multimodal immersive games combine multimodal interaction technology with games to provide learners with an authentic, interesting, and challenging learning environment [3]. In the game, learners can not only deeply perceive and understand the cultural context but also experience the process of problem solving and improve their abilities and qualities through practical operation.

In recent years, games developed based on experiential learning theories have gained widespread attention in the field of education due to their context-awareness advantages, but they still suffer from a slight lack of interactivity and fun [15] as well as cultural immersion [11]. Meanwhile, the current C-STEAM education is still in its infancy and relies on the traditional course teaching method in terms of situational awareness, which is able to transfer knowledge but has limitations in stimulating students' interest and promoting their active exploration and interaction [9]. Therefore, this study will combine experiential learning and the "6C" teaching model of C-STEAM programs [18], take Yangjiang Kites, one of the intangible cultural heritages, as an example, to design a multimodal immersive games for C-STEAM programs, and apply it in C-STEAM teaching to promote students' learning experience and learning effect. This study aims to address: (1) How to design multimodal immersive games for C-STEAM courses? (2) How can multimodal immersive games be applied to C-STEAM programs? (3) How effective can multimodal immersive games be in C-STEAM programs?

2 THE DESIGN AND DEVELOPMENT OF MULTIMODAL IMMERSIVE GAMES FOR C-STEAM PROGRAMS

2.1 The design of multimodal immersive games for C-STEAM programs

Experiential learning is a learning process that involves gaining direct experience through hands-on activities and observing, reflecting on, and mentally substituting the activities to acquire knowledge and skills. It consists of four stages: concrete experience, observation and reflection, abstract conceptualization, and active experimentation [10]. Concrete experience refers to learners' activities in real or simulated situations to acquire knowledge and skills, observation and reflection refers to learners' reflective assessment and analysis in the process of experience, abstract conceptualization refers to learners' sorting and summarizing the perceived knowledge to establish a structured memory system, and active experimentation refers to the practical application of the acquired knowledge and skills by learners.

Many current studies have been conducted to design multimodal immersive game models based on experiential learning theory. One of the most famous models is the experiential gaming model proposed by Killi [6], which constitutes the cycle of the experiential learning process with four parts: problem challenge, active experimentation, reflective observation, and schemata construction. The model emphasizes the promotion of in-depth understanding and application of knowledge through practical learning, reflection, and conceptual construction. In the learning of culture, creating authentic questioning scenarios based on cultural themes and understanding the cultural connotation of the theme step by step through basic reading, cultural observation, and skill training is the key to prompting students to form cultural deposits at the emotional and cognitive levels [18].

Therefore, based on the "6C" teaching model of C-STEAM, this study constructs a multimodal immersive game model for C-STEAM programs in combination with experiential learning theory (Figure 1). That is, games need to create immersive cultural scenarios in the context of specific cultural themes and guide learners to solve specific problems to enhance the fun and playability of the game by setting up rich and culturally specific interaction mechanisms so that the learners can complete the challenges through embodied interactions and learn about the cultural background and craftsmanship techniques; to set up a feedback mechanism during the task to supplement and expand the materials related to the theme culture, so that the learners can understand the cultural knowledge and at the same time guide the learners to make reflective observation; to help learners build a structured memory system by refining cultural features after the task is completed, such as through the use of knowledge cards.

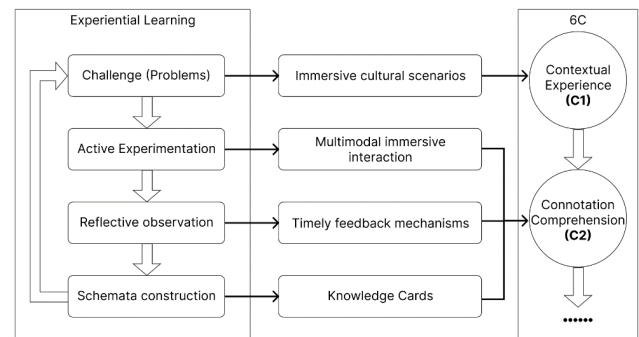


Figure 1: The multimodal immersive game model for the C-STEAM programs

2.2 The design of multimodal immersive games for C-STEAM programs

Relying on the multimodal immersive game model for the C-STEAM programs, this study developed a Web-based kite-making educational game, "Yangjiang Kite Journey" based on Cocos Creator and Touch Designer, taking Yangjiang Kite as an example of non-heritage. Yangjiang Kite is one of the representatives of the "Southern Harrier" in the Chinese kite culture "Southern Yao and Northern Yuan Kites", which is rich in aesthetic features, craft value, and

folklore meaning. At present, the development of the C-STEAM program with the theme of Yangjiang Kites is still in the exploratory stage, and it is mainly carried out in the form of graphic explanation and skill experience.

Based on this, this study will take C1 and C2 of the "6C" teaching model as the design requirements of the game: (1) Contextual Experience(C1): create an immersive scene for students to understand the cultural background, aesthetic characteristics, and folklore of Yangjiang Kites; (2) Connotation Comprehension(C2): provide multimodal interactions and feedbacks, so that students can deeply participate in the cultural experience, and learn the process of making Yangjiang Kites. Referring to the structure of the game model in this study, the content of "Yangjiang Kite Journey" mainly includes:

2.2.1 Challenge (Problems). The game presents challenging tasks through narrative design to guide learners to participate in the game and gain concrete experiences. Specifically, the game guides learners to participate in the kite competition, the Yangjiang Kite Folklore Activity, in the form of invitations (Figure 2), in which learners are required to complete the production and flying of kites according to the four stages of the Yangjiang Kite production process, namely, "tying, gluing, painting, and flying". The interactive narrative structure of the game is realized through the scene switching in Cocos Creator. The game has set up the "introduction", "skeleton tying", "kite gluing", "kite painting" and "kite flying", which are switched sequentially following the timeline of the learner's play. To enhance the immersion of the game, visually, the game creates real kite-making scenes and designs classic kite styles such as "flowers, birds, fish, and insects" following the physical kites (Figure 3); aurally, the game carries out music and sound effect design following the background of the story and the development of the narrative plot in a consistent style.



Figure 2: Game UI - Kite Race Invitation

2.2.2 Active Experimentation. In the four production stages of Yangjiang Kites, namely "skeleton tying", "kite gluing", "pattern painting" and "kite flying", the game designs corresponding interactive tasks according to the characteristics of the techniques so that the students can deepen their understanding of the culture through multimodal interaction in practice. For example, students can assemble sticks to tie the skeleton, paint the kite to make paper and customize the kite with different patterns and colors (Figures 4–6). In the "kite flying" stage, the game is designed for different usage scenarios, including online Web interaction and offline device interaction (Figures 7–8). For the general classroom experience, the learner can place the kite made in the online kite flying scene and

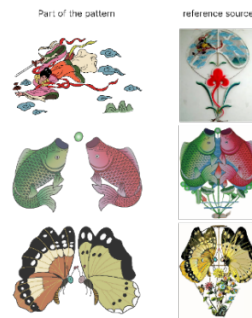


Figure 3: Part of the pattern reference source

complete the kite flying by blowing into the microphone peripheral to provide power for the kite. For the venue display experience, learners can transfer the kite made online to the offline interactive device developed by Touch Designer through a WebSocket connection and complete the kite flying through the interactive device (anemometer and spools).

2.2.3 Reflective Observation. Before each task, the game provides learners with an introduction to the craft so that they can understand the characteristics of the kite-making process and related knowledge. During the task, the game provides a gameplay introduction for each interactive task and sets up corresponding success feedback and failure feedback. For learners who failed the task or were unable to complete the task for a long time, the game triggered a failure analysis or gameplay hints to reduce the difficulty of the task (Figures 9–10).

2.2.4 Schemata Construction. When the tasks are completed, visualized knowledge cards summarize the related conceptual knowledge (Figures 11–12), including the introduction of skeleton styles, the analysis of patterns' symbolism, the precautions to be taken for kite flying, and so on. This further promotes the learners' overall conceptual construction of the knowledge of Yangjiang Kites.

3 THE APPLICATION OF MULTIMODAL IMMERSIVE GAMES IN THE C-STEAM PROGRAM

In the "6C" teaching model, multimodal immersive games will be used as teaching resources to help students gain a deeper understanding of cultural characteristics and form cultural deposits in the Contextual Experience and Connotation Comprehension phases, so as to pave the way for subsequent practical learning. In this study, the educational game "Yangjiang Kite Journey" will lead students to understand the process of making Yangjiang Kites and the cultural meaning behind them through the story of the kite competition. Specifically, in the Yangjiang Kite C-STEAM program:

In the Contextual Experience (C1) and Connotation Comprehension (C2) phases, the teacher first gives a general introduction to Yangjiang Kites and then guides students to play the "Yangjiang Kite Journey" kite game through PC or mobile devices. Students complete the challenges step by step through the game, understand the connotation of kite culture in the immersive game, and form a preliminary understanding of the process of kite making. At the



Figures 4–6: The interface of the game interaction



Figures 7–8: Scenarios for the use of "Kite Flying": general classroom (left), venue display (right)



Figures 9–10: The interface of the game interaction feedback



Figures 11–12: The interface of the knowledge card

end of the game, the teacher guides the students to express their feelings and discuss what they have learned and experienced during the game, further deepening their understanding of and interest in kite culture.

In the stages of Characteristic Inquiry (C3), Create Artifact (C4) and Connect Society (C5), with the help of local resources in Yangjiang, the teacher leads students to visit the Yangjiang Kite

Museum and the Yangjiang Kite Workshop, guiding them to conduct further research and investigation on cultural characteristics. During the workshop, students will participate in the production of physical kites, design and create their own kite works, and present and share their works and production experience after the works are completed. Finally, the teacher will lead the students to fly their kites outdoors.

Finally, in the Conclusive Reflection (C6) stage, students' performance in the Yangjiang Kite C-STEAM course is evaluated as a whole through the behavior of kite game operation, test of basic knowledge of kites, feedback of classroom experience, and mutual evaluation of kite works. In this way, students are encouraged to rethink the connotation of Yangjiang Kite culture, develop a deeper analysis, and reconstruct cultural understanding.

4 THE APPLICATION OF MULTIMODAL IMMERSIVE GAMES IN THE C-STEAM PROGRAM

4.1 Participants and procedure

This quasi-experimental study investigates the impact of multimodal immersive games on student learning outcomes in a C-STEAM program. In this study, two classes of students totaling 85 students, 43 in the experimental class and 42 in the control class, were selected from the senior class of a middle school in Yangjiang City, Guangdong Province, China. The results of the a priori survey showed that the students in both classes had not been exposed to Yangjiang Kite. In the experiment, students in both classes were asked to participate in the C-STEAM program of Yangjiang Kites, in which the experimental class introduced the game "Yangjiang Kite Journey" to assist teachers in the C-STEAM program of Yangjiang Kites. The control class followed the normal C-STEAM program.

Before the lesson, students were assessed on their knowledge of Yangjiang Kite culture. Afterward, the teacher taught according to the "6C" teaching model. In the stage of Contextual Experience (C1) and Connotation Comprehension (C2), for the experimental group class, the teacher would use the game "Yangjiang Kite Journey" as an introduction to guide the students to experience the kite game to learn about the process of making Yangjiang Kites and the cultural background of Yangjiang Kites. Then, the teacher would supplement the explanation to lead the students to study further and explore the knowledge. In the control group, students were introduced to the class by watching a video. Afterward, both classes were required to complete the C-STEAM curriculum learning of Characteristic Inquiry (C3), Create Artifact (C4), Connect Society (C5), and Conclusive Reflection (C6). At the end of the study, students in both the experimental and control groups filled out questionnaires, and a sample of students from both groups were interviewed for about 5 minutes each.



Figure 13: The application of multimodal immersive games in the C-STEAM program

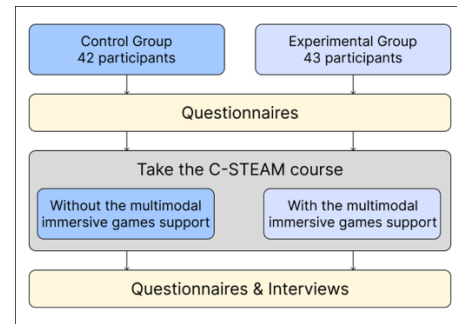


Figure 14: Experimental Procedure

4.2 Measuring instruments

To validate the effectiveness of multimodal immersive games for C-STEAM programs, this study used a mixed-method approach of questionnaires and semi-structured interviews to assess the learning quality of applying the "Yangjiang Kite Journey" educational game. In the questionnaire section, the study designed a questionnaire based on the evaluation dimensions of experiential educational games for ICH in handicrafts proposed by Zhu et al. [5] and combined with the content of this study, which included three dimensions: learning experience, learning effect, and learning motivation. Learning experiences are the emotional responses and cognitive feelings that students experience after engaging in learning, which include a sense of immersion, a level of enjoyment, a sense of control over the game, and a level of understanding. Learning effect refers to the degree of students' mastery of the knowledge of non-heritage skills in the learning process, which in this study involves kite styles, craftsmanship techniques, cultural contexts, historical backgrounds, and so on. Learning motivation to learn is the motivation of students to learn in a C-STEAM program that incorporates multimodal immersive games, which includes a willingness to participate in the game-based learning approach, a willingness to learn about ICH as a whole, and a willingness to identify culturally with what they are learning.

In this study, the scale was presented as a Likert scale, and participants were asked to rate a pair of items from five options, ranging from "Strongly Disagree" to "Strongly Agree". Also, to further validate the applicability of the scale to this study, 30 students from another class in the school were selected for a pilot study to test the reliability of the scale before the start of this study. The results showed that the KMO values of the three indicators of learning experience, learning effect, and learning motivation were all greater than 0.6, the cumulative squared deviation interpretations were all greater than 60%, and the Cronbach's alpha values were all greater than 0.8, which indicated that the reliability and validity of the scales were good overall.

In addition, semi-structured interviews were used in this study to understand further learners' feelings about game learning and to provide explanations, confirmations, and additions to the questionnaire results. The interviews were conducted as face-to-face audio recordings to collect feedback on learners' satisfaction with learning, including game experience, sense of fun, usefulness, and knowledge acquisition preferences. After the interviews, the recordings'

Table 1: The results of the Mann-Whitney U test analysis

Dimensions	Items	Mean value (P25~P75)		Mann-Whitney			p
		control group	experimental group	U	W	Z	
Learning	Q1	5.00(4.00~5.00)	5.00(5.00~5.00)	706.500	1609.500	-2.381	0.017*
Experience	Q2	5.00(4.00~5.00)	5.00(5.00~5.00)	776.500	1679.500	-1.635	0.003**
	Q3	5.00(4.00~5.00)	5.00(5.00~5.00)	682.000	1585.000	-2.859	0.004**
	Q4	5.00(4.00~5.00)	5.00(5.00~5.00)	816.000	1719.000	-1.127	0.022*
Learning Effect	Q5	5.00(4.00~5.00)	5.00(5.00~5.00)	686.000	1589.000	-2.966	0.003**
	Q6	5.00(4.00~5.00)	5.00(5.00~5.00)	703.500	1606.500	-2.524	0.012*
	Q7	5.00(4.00~5.00)	5.00(5.00~5.00)	682.000	1585.000	-2.859	0.004**
	Q8	5.00(4.00~5.00)	5.00(5.00~5.00)	646.500	1549.500	-3.170	0.002**
	Q9	5.00(4.00~5.00)	5.00(5.00~5.00)	705.000	1608.000	-2.630	0.009**
Learning	Q10	5.00(4.00~5.00)	5.00(5.00~5.00)	700.500	1603.500	-2.684	0.007**
Motivation	Q11	5.00(4.00~5.00)	5.00(5.00~5.00)	770.000	1673.000	-1.818	0.035*
	Q12	5.00(4.00~5.00)	5.00(5.00~5.00)	747.500	1650.500	-2.061	0.039*

results were converted into text form and organized for subsequent analysis.

5 RESULT

5.1 Data analysis of questionnaires

According to the normality test performed on the data of the questionnaires of the two groups of students, it is clear that the experimental data do not conform to normal distribution ($p < 0.05$). Therefore, this study used a non-parametric test to analyze the data. Due to the small sample size of data involved in this study, the experimental data were analyzed using the Mann-Whitney U test, and the results are shown in Table 1.

By comparing the asymptotic significance, it can be found that there is a significant difference between the experimental group and the control group in terms of overall learning outcomes, learning experience, and motivation after the course ($p < 0.05$). In terms of median, although the median of both the experimental and control groups is 5.00, it can be found by comparing the quartiles that the overall results of the experimental group are more centralized, which can be concluded that the multimodal immersive game has significantly improved the quality of students' learning in the C-STEAM program. In terms of learning experience, compared with simple explanations, games with rich interactions are more likely to stimulate students' interest in learning and give them a sense of immersion in the learning process. In terms of learning effect, most of the questions in this dimension showed highly significant differences ($p < 0.01$), which shows that compared with traditional illustrations and texts, multimodal immersive games can better promote students' mastery of cultural knowledge and understanding of cultural connotations. Regarding learning motivation, the students who passed the multimodal immersive games also had a better sense of cultural identity and a better willingness to follow up learning than those in the control group.

5.2 Data analysis of interviews

By organizing and analyzing the audio-recorded texts of students' semi-structured interviews, this study found that:

First of all, the description of the learning experience of Yangjiang Kite flying by the students in the experimental group mainly focuses on the two aspects of learning mode and learning fun. The students in the experimental group generally believe that multimodal immersive games are more attractive than traditional learning modes, and Liu believes that "Compared with the direct lectures, the multimodal immersive games are more capable of concentrating my energy and improving my hands-on practical ability." Zhang exclaimed, "In the process of Yangjiang Kite 'painting', I can freely combine my favorite patterns, and each combination will have a different meaning, which is really interesting!"

Second, the experimental group students' descriptions of the learning effect of Yangjiang Kites mainly focused on the degree of understanding of Yangjiang Kites, specifically covering its historical research, cultural value, production process, and precautions. More than 71% of the students in the experimental group mentioned that the Yangjiang Kite game had left a deep impression on them. Luo said, "In the process of drawing kite patterns, it deepened my understanding of the meanings of different patterns, such as those of fish, insect, flower, and other kinds of motifs," and Wang lamented, "Yangjiang Kites have a long history and have been made the hometown of national kites and listed in the list of intangible cultural heritage, I experienced the process of making Yangjiang Kites and the historical inheritance in this game, which improved my hands-on practical ability and awareness of cultural heritage protection."

Finally, the description of the experimental group students' motivation to learn Yangjiang Kites mainly focused on the cultural heritage of Yangjiang Kites, and most of the students believed that both multimodal immersive and traditional learning methods enhanced their interest in learning intangible cultural heritage, but multimodal immersive was more motivating. Meanwhile, 90% of the students indicated that they would like to continue to use multimodal immersive games to learn other intangible cultural heritage cultures in the future. In terms of suggestions for improvement of multimodal immersive games, Li said, "The learning content can be further enriched so that I can learn more knowledge about intangible cultural heritage culture in the process of the game, and the

difficulty and challenge of the game can be appropriately increased at the same time.”

This interview shows that multimodal immersive games have significant advantages in the transmission of intangible cultural heritage culture and can enhance learners’ learning effect, learning experience, and learning motivation.

6 DISCUSSION

This study proposes a model of multimodal immersive games for C-STEAM courses and develops corresponding educational game using Yangjiang Kite as an example. In the Yangjiang Kite C-STEAM program, the multimodal immersive games significantly affected students’ learning experience, learning effect, and learning motivation.

In the dimension of learning experience, the multimodal immersive game designed in this study creates a game scenario by combining Yangjiang Kite folklore activities and provides students with clear challenge goals and feedback, which to some extent facilitated students’ learning experience of cultural learning. As Kiili argues, clear goals, clear feedback, and other game design elements are the key to enhancing students’ mind-flow experience [7]. However, from the results of the study, the analysis results of questionnaire Q1, “I can get immersive experience during the game” did not reach a highly significant effect, which may be because the game is not well-designed in terms of audio-visual elements. Paras’s research has pointed out that aesthetically pleasing audio-visual elements in instructional games can give students a better sense of immersion in the game [12]. For learning effect, the educational game developed in this study not only designs rich multimodal interaction and feedback mechanisms, but also interspersed the game with a variety of cultural knowledge science, which significantly enhanced the students’ cognitive level of Yangjiang Kites. This further confirms Alexiou et al.’s observation that timely and frequent feedback can enhance users’ cognitive engagement when introduced in a non-intrusive way [1]. In addition, the integration of multimodal immersive game into C-STEAM programs in this study significantly increased students’ learning motivation, which has been found to have similar results in past studies. Patrick et al. explored changes in students’ motivation through a gamification intervention, noting that gamified learning can have a positive impact on students’ motivation and engagement [2]. Overall, through the multimodal immersive game students not just by passively accepting knowledge, but by actively exploring and learning through the game’s interactions and challenges, and this proactivity significantly increases students’ interest in learning about the culture, which further motivates them to explore the subsequent content.

7 CONCLUSION AND FUTURE STUDY

Aiming at the problems existing in the current C-STEAM programs teaching and combining with the characteristics of experiential learning, this study constructs a model of multimodal immersive games for C-STEAM programs and the basic mode of its application in C-STEAM programs. On this basis, this study designs “Yangjiang Kite Journey” games and uses it in the classroom. The results of the study show that the multimodal immersive games

designed in this study have a positive effect on learners’ learning effect, learning experience, and learning motivation in the C-STEAM program. Compared with traditional teaching, first-person participatory experiences can deepen students’ understanding of cultural connotations. At the same time, the multimodal interactive tasks in the game also increase the fun of learning, better stimulate learners’ interest in cultural learning, and lay the foundation for subsequent cultural practice.

However, there are still some issues in the game design of this study, such as the shallow depth of game cultural background knowledge and the current shortcomings in the game artwork design. In the future, “Yangjiang Kite Journey” will further enrich the game content, promote the combination of cultural background knowledge and game tasks, improve the game details, enhance the game immersion, and better apply it in the C-STEAM program.

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